

Cyberon TTS gRPC Protocol

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Leading Speech Solution provider

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1.0.1	2022/04/14	Cyberon	Add sample code
1.0.2	2025/01/24	Cyberon	Add Chinese surname tag description

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1. gRPC 連線資訊

Field	Value
IP	TTS entry IP
PORT	TTS entry PORT

2. service.proto

2.1 TtsRequest

Field	Type	Value	Description
serviceName	string	e2e	必要，服務名稱
text	string		必要，合成文字
outfmt	string	wav	輸出聲音格式，預設 mp3，支援 mp3, wav, pcm(16K), pcm8(8K)
vbr_quality	int64	4	outfmt=mp3 時，音質設定，預設 4，1(好) ~ 9(差)
language	string	zh-TW	主要語言，預設：zh-TW
phrbrk	bool	false	是否自動斷詞，預設：false
speaker	string	Sharon	主要語言語者
speed	float	1	主要語言語速，預設 1，範圍 0.5 ~ 2
gain	float	1	主要語言聲音大小，預設 1，範圍 0.5 ~ 4
token	string	...	必要，賽微提供
uid	string	...	user unique id

2.1.1 中文姓氏語音合成標記

1. 說明

由於中文姓氏在進行語音合成時，許多姓氏會遇到破音字的問題，例如：曾、仇、單等等...。賽微語音合成服務提供中文姓氏標記的支援，協助您將姓氏合成為正確的讀音。

2. 標記方式

請在透過 `text` 參數傳遞合成文字給賽微語音合成服務時，使用 `[name]` 及 `[/name]` 將姓名文字置入標記中，如此賽微的語音合成的服務將會以正確的中文姓氏唸法為您合成聲音。

3. 標記範例

`[name]` 曾經國 `[/name]`

`[name]` 仇曼青 `[/name]`

2.2 TtsResponse

Field	Type	Label	Description
data	bytes		

2.3 StreamService

Method Name	Request Type	Response Type	Description
TTS	TtsRequest	TtsResponse stream	

3. Scalar Value Types

.proto Type	Notes	C++	Java	Python	Go	C#	PHP	Ruby
double		double	double	float	float64	double	float	Float
float		float	float	float	float32	float	float	Float
int32	Uses variable-length encoding. Inefficient for encoding negative numbers – if your field is likely to have negative values, use sint32 instead.	int32	int	int	int32	int	integer	Bignum or Fixnum (as required)
int64	Uses variable-length encoding. Inefficient for encoding negative numbers – if your field is likely to have negative values, use sint64 instead.	int64	long	int/long	int64	long	integer/string	Bignum
uint32	Uses variable-length encoding.	uint32	int	int/long	uint32	uint	integer	Bignum or Fixnum (as required)
uint64	Uses variable-	uint64	long	int/long	uint64	ulong	integer/string	Bignum or Fixnum (as required)

.proto Type	Notes	C++	Java	Python	Go	C#	PHP	Ruby
	length encoding.							
sint32	Uses variable-length encoding. Signed int value. These more efficiently encode negative numbers than regular int32s.	int32	int	int	int32	int	integer	Bignum or Fixnum (as required)
sint64	Uses variable-length encoding. Signed int value. These more efficiently encode negative numbers than regular int64s.	int64	long	int/long	int64	long	integer/string	Bignum
fixed32	Always four bytes. More efficient than uint32 if values are often greater than 2 ²⁸ .	uint32	int	int	uint32	uint	integer	Bignum or Fixnum (as required)
fixed64	Always eight bytes. More efficient than	uint64	long	int/long	uint64	ulong	integer/string	Bignum

.proto Type	Notes	C++	Java	Python	Go	C#	PHP	Ruby
	uint64 if values are often greater than 2^56.							
sfixed32	Always four bytes.	int32	int	int	int32	int	integer	Bignum or Fixnum (as required)
sfixed64	Always eight bytes.	int64	long	int/long	int64	long	integer/string	Bignum
bool		bool	boolean	boolean	bool	bool	boolean	TrueClass/FalseClass
string	A string must always contain UTF-8 encoded or 7-bit ASCII text.	string	String	str/unicode	string	string	string	String (UTF-8)
bytes	May contain any arbitrary sequence of bytes.	string	ByteString	str	[]byte	ByteString	string	String (ASCII-8BIT)

4. gRPC client sample code

4.1 說明

1. 本範例以 Golang 為例
2. 下列資訊將由賽微提供，以利您開發串接程式 (若您缺少下列資訊，請洽詢賽微窗口)
 - i. TTS gRPC entry point (IP, port)
 - ii. Token
 - iii. service.proto 檔案 (Protocol Buffers 檔)

4.2 gRPC client 開發注意事項

1. 請先將 proto 檔轉檔為您所使用的程式語言所支援的 pb 檔
2. 請確認您的執行環境是否能與 gRPC server 正確連線

4.3 Client 端開發範例程式

1. 前提

本範例所連線的 server 環境為測試環境，使用自簽 SSL 憑證，故需略過憑證安全性檢查，請依據您的環境調整此部份設定

2. 程式碼範例

```

package main

import (
    "context"
    "crypto/tls"
    "fmt"
    "io"
    "log"

    /* *** Please import your pb file here. *** */

    "google.golang.org/grpc"
    "google.golang.org/grpc/credentials"
)

func main() {
    //Bypass SSL certificate secure verify
    config := &tls.Config{
        InsecureSkipVerify: true,
    }
    conn, err := grpc.Dial("[gRPC_server_IP]:[gRPC_service_port]",
        grpc.WithTransportCredentials(credentials.NewTLS(config)))
    if err != nil {
        panic(fmt.Sprintf("dial grpc server error: %v", err))
    }
    defer conn.Close()
    client := protos.NewStreamServiceClient(conn)
    err = printLists(client, &protos.TtsRequest{
        ServiceName: "[service_name]",
        Text:         "今天天氣真好",
        Outfmt:       "wav",
        Language:     "zh-TW",
        Speaker:      "Sharon",
        Token:        "[your_service_token]",
        Uid:          "[user_unique_id]"}
    )
    if err != nil {
        log.Fatalf("printLists.err: %v", err)
    }
    log.Println("end...")
}

func printLists(client protos.StreamServiceClient, r *protos.TtsRequest) error {
    stream, err := client.TTS(context.Background(), r)
    if err != nil {
        return err
    }
    for {
        resp, err := stream.Recv()
        if err == io.EOF {
            break
        }
        if err != nil {
            return err
        }
        log.Println("resp: ", string(resp.Data))
        /* Process your response data here. */
    }
    return nil
}

```